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On the Theoretical Understanding of Identifiable Sparse Autoencoders and Beyond

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Abstract

Sparse autoencoders (SAEs) have emerged as a powerful tool for interpreting features learned by large language models (LLMs). It aims to recover complex superposed polysemantic features into interpretable monosemantic ones through feature reconstruction via sparsely activated neural networks. Despite the wide applications of SAEs, it remains unclear under what conditions an SAE can fully recover the ground truth monosemantic features from the superposed polysemantic ones. In this paper, through theoretical analysis, we for the first time propose the necessary and sufficient conditions for identifiable SAEs (SAEs that learn unique and ground truth monosemantic features), including 1) extreme sparsity of the ground truth feature, 2) sparse activation of SAEs, and 3) enough hidden dimensions of SAEs. Moreover, when the identifiable conditions are not fully met, we propose a reweighting strategy to improve the identifiability. Specifically, following the theoretically suggested weight selection principle, we prove that the gap between the loss functions of SAE reconstruction and monosemantic feature reconstruction can be narrowed, so that the reweighted SAEs have better reconstruction of the ground truth monosemantic features than the uniformly weighted ones. In experiments, we validate our theoretical findings and show that our weighted SAE significantly improves feature monosemanticity and interpretability.

1 Introduction

The "black-box" nature is a long-standing problem plaguing the mechanistic interpretability of deep neural networks. One of the key problems is feature *polysemanticity*, where an individual neuron is often activated by multiple semantically unrelated features [6, 9, 20, 26]. This issue is particularly evident in large language models (LLMs) as neurons rarely correspond to distinct and well-defined features. Previous studies have proposed the *superposition* hypothesis that the polysemantic features are linear combinations of the *monosemantic* ones, so that models can represent more features than they have dimensions [6, 9, 20].

To disentangle the superposed polysemantic features, sparse autoencoders (SAEs) [8, 14, 15, 16] have demonstrated significant potential for identifying interpretable monosemantic features. They have been widely used as a promising approach to enhancing the interpretability of LLMs [4, 7, 8, 12] and VLMs [5, 13]. Typically, an SAE has a single wide layered encoder-decoder architecture, with sparse activations such as ReLU [4] or Top-k [8]. While previous works mainly focused on

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studying architecture [2, 14, 18, 21] or evaluation [15] of SAEs, the theoretical understanding of the identifiability of SAEs is still lacking. Specifically, we wonder: *Can SAEs uniquely recover the ground truth monosemantic features from the polysemantic inputs?*

To answer the above question, in this paper, we investigate the identifiability of SAEs. Specifically, based on the polysemantic features learned by a single-layer ReLU activated toy model, we derive the closed-form optimal solution to SAEs under the extreme sparsity of the ground truth monosemantic features. By proving that the optimal solution is unique and precisely recovers the ground truth features, we show the sufficient conditions for identifiable SAEs (extreme sparsity of ground truth features, sparse activation, and enough hidden dimensions). Besides, through several counterexamples, we show that the sufficient conditions are also necessary. Moreover, when the extreme sparsity assumption does not meet (SAE unidentifiable), we propose a reweighting strategy to enhance the reconstruction of the ground truth features, and propose the principle for the weight selection through theoretical analysis. Specifically, we derive the theoretical relationship between the loss of SAE feature reconstruction and that of the ground truth feature reconstruction, and discuss that smaller weights on the more polysemantic dimensions help reduce the negative inferences hindering the reconstruction of ground truth features.

Our contributions are summarized as follows.

- We for the first time propose the necessary and sufficient conditions for identifiable SAEs to recover the ground truth monosemantic features from the superposed polysemantic ones. The conditions include 1) extreme sparsity of the ground truth feature x, 2) sparse activation of SAEs, and 3) enough hidden dimensions of SAEs.
- When the conditions for identifiable SAEs do not meet, we propose a reweighting strategy to improve the reconstruction of ground truth features by SAEs and theoretically discuss the principle of weight selection.
- We validate our theoretical findings through experiments and show that our reweighting strategy significantly enhances the monosemanticity and interperability of SAE features.

2 Related Works

2.1 Polysemanticity and Superposition Hypothesis.

A widely accepted explanation for feature polysemanticity is the superposition hypothesis [6, 9, 20], which regards a polysemantic dimension to be an approximately linear combination of several natural semantic concepts. To investigate the mechanism of polysemanticity, mainstream studies reveal that superposition occurs when a model represent more features than they have dimensions. Specifically, [6] introduced a single-layer toy model demonstrating that polysemanticity occurs when ReLU networks are trained on synthetic data with sparse input features. [9] proved that superposition is actively helpful for efficiently accomplishing the task of emulating circuits via mathematical models of computation. [20] explained polysemanticity through the lens of feature capacity, suggesting that the optimal capacity allocation tends to polysemantically represent less important features. Aside from the dimension restriction, [10] demonstrated that polysemanticity can also incidentally occur because of sparse regularization or neural noise.

In this paper, we adopt the superposition hypothesis to simulate the generation of polysemantic features from the ground truth monosemantic inputs.

2.2 Sparse Autoencoders (SAEs).

Sparse autoencoder (SAEs) automatically learn features from unlabeled data, typically trained with sparsity priors such as ReLU activation and l-1 penalty [16]. To mitigate the feature suppression caused by l-1 regularization [21, 25], [14] proposed k-sparse autoencoders, which replace the l-1 penalty with a Top-k activation function. To address the same problem, [18] proposed the gated SAE that decouples detection of which features are present from estimating their magnitudes. Besides, to ensure the features learned are functionally important, [2] propose to optimize the downstream KL divergence instead of reconstruction MSE. Moreover, by using k-sparse autoencoders, [8] found clean scaling laws with respect to autoencoder size and sparsity. [15] proposed a suite of evaluations for SAEs to analyze the quality of monosemantic features by focusing on polysemous words. They

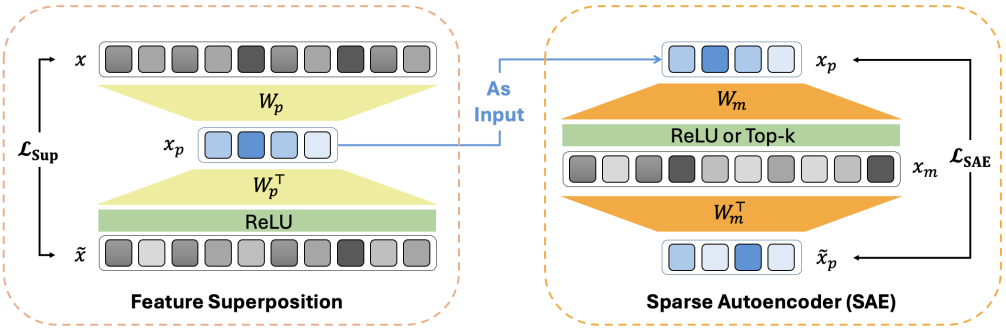


Figure 1: Mathematical modeling of feature superposition and sparse autoencoder (SAE). The superposed polysemantic features x_p learned by the toy neural network model serve as the input to the SAE. For an identifiable SAE, x_m should uniquely recover the ground truth x.

showed that compared with alternative activations such as JumpReLU and Top-k, ReLU still leads to competitive semantic quality.

In this paper, we investigate the identifiability of SAEs. Specifically, we are interested in the conditions under which SAEs can uniquely recover the ground truth monosemantic features from the polysemantic inputs. We consider the reconstruction-based SAEs with ReLU or Top-k activations.

3 Necessary and Sufficient Conditions for Identifiable SAEs

In Section 3.1, we first present the mathematical formulations including a toy model that transforms ground truth monosemantic features into superposed polysemantic ones, and an SAE aiming to recover the ground truth features from the polysemantic inputs, illustrated in Figure 1. In Sections 3.2.1 and 3.2.2, we discuss the sufficient and necessary conditions for identifiable SAEs, respectively. All proofs can be found in Appendix A.1.

3.1 Mathematical Formulations

Notations. Denote $x := (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$ as the ground truth monosemantic feature with dimension $n > 0$, and $x_p \in \mathbb{R}^{n_p}$ as the superposed polysemantic feature, where we assume $n > n_p > 0$. Moreover, we use $x_m \in \mathbb{R}^{n_m}$ to denote the feature learned by sparse autoencoder, where $n_m > n_p$. For the weight matrices, we denote $W_p \in \mathbb{R}^{n_p \times n}$ and $W_m \in \mathbb{R}^{n_m \times n_p}$. Throughout this paper, $\|\cdot\|$ denotes the l_2 norm if not specified. For mathematical conciseness, we denote $\mathbf{1}$ as the all-one vector, and $\mathbf{0}$ as the all-zero vector. Also, we denote $[n] = \{1, \ldots, n\}$. Furthermore, for a matrix W we use $W_{[i,:]}$ to denote the i-th row of W, and use $W_{[:,j]}$ to denote the j-th column of W. For a vector x, we use x_i to denote the i-th element of x.

Sparsity of ground truth features. Following [6], we assume that each dimension of the ground truth features is non-negative, i.i.d., and sparsely distributed. Specifically, under a sparse factor $S \in [0, 1]$, for $i \in [n]$, we have $x_i = 0$. Larger S indicates higher feature sparsity.

Superposition of features. We formulate the superposed polysemantic feature as a linear transformation of the ground truth monosemantic input feature x. Following the toy model in [6], we assume the polysemantic features are trained based on a ReLU activated output model where

$$x_p = W_p x \tag{1}$$

$$\tilde{x} = \text{ReLU}(W_p^\top x_p). \tag{2}$$

The learned features are $x_p = W_p^* x$ with the optimal weight matrix W_p^* learned by minimizing the reconstruction loss function

$$\mathcal{L}_{\text{Sup}}(W_p; x) = \mathbb{E}_{x_p}\|x - \tilde{x}\|^2 = \mathbb{E}_x\|x - \text{ReLU}(W_p^\top W_p x)\|^2. \tag{3}$$

Sparse autoencoders (SAEs). We use a sparse autoencoder to recover the ground truth monosemantic feature x from the polysemantic input x_p. SAE is a neural network with a single hidden layer and

a sparse activation function $\sigma : \mathbb{R}^{n_m} \to \mathbb{R}^{n_m}$ such as ReLU [4] or Top-k [8]. For mathematical simplicity, we omit the bias term and define the encoder and decoder as

$$x_m = \sigma(W_m x_p) \tag{4}$$

$$\tilde{x}_p = W_m^\top x_m. \tag{5}$$

It is trained with a reconstruction loss

$$\mathcal{L}_{\text{SAE}}(W_m; x_p) = \mathbb{E}_{x_p}\|x_p - \tilde{x}_p\|^2 = \mathbb{E}_x\|W_p x - W_m^\top \sigma(W_m W_p x)\|^2. \tag{6}$$

As not all SAE architectures include the sparsity regularization, e.g. [8], we here omit the l_1 penalty and focus on the reconstruction loss only. Then under the optimal weight matrix

$$W_m^* = \arg\min_{W_m} \mathcal{L}_{\text{SAE}}(W_m; x_p) \tag{7}$$

the learned monosemantic feature is $x_m = \sigma(W_m^* x_p)$. Ideally, we expect x_m to fully recover x, i.e., $x_m = x$ if $n_m = n$ and $x_m = (x, \mathbf{0})^\top$ if $n_m > n$.

Identifiability. Identifiability refers to the property that the parameters of a model can be uniquely determined from the observed data [3, 11, 19, 27]. Typically, a model p_θ is identifiable if for all $\theta, \theta' \in \Theta$, we have

$$p_\theta = p_{\theta'} \;\Rightarrow\; \theta = \theta'. \tag{8}$$

In this paper, we say an SAE is identifiable if the reconstruction of polysemantic features x_p leads to the perfect reconstruction of ground truth feature x, i.e.,

$$\tilde{x}_p = x_p \;\Rightarrow\; \tilde{x} = x. \tag{9}$$

Or equivalently, from the perspective of the loss function, we say an SAE is identifiable if the solution $W_m^* = \arg\min_{W_m} \mathcal{L}_{\text{SAE}}(W_m; x_p)$ is unique and recovers the ground truth monosemantic features, i.e., $\sigma(W_m^* x_p) = x$.

3.2 Necessary and Sufficient Conditions for Identifiable SAE

In this section, we derive the necessary and sufficient conditions of SAE fully recovering the ground truth monosemantic feature x. These conditions include: 1) extreme sparsity of the ground truth feature x, 2) sparse activation of SAE, and 3) enough hidden dimensions of SAE. It is worth noting that although widely used in SAE methods, sparsity regularization such as l_1 norm is not necessary.

3.2.1 Sufficient Conditions for SAE Full Recovery

First of all, in Theorem 1, we find a solution to SAEs that can fully recover the ground truth feature x.

Theorem 1 (Optimality)**.** *Let $\mathcal{L}_{\text{SAE}}$ be defined in Equation* (6) *with sparse activation function σ. If $n_m \geq n$ and $S \to 1$, then we have $W_m^* = (W_p, \mathbf{0})^\top \in \arg\min_{W_m} \mathcal{L}_{\text{SAE}}(W_m; x_p)$.*

Theorem 1 shows that when the ground truth feature x is extremely sparse and the hidden dimension n_m is large enough, $W_m^* = (W_p, \mathbf{0})^\top$ is the solution to minimizing the SAE loss function. As follows, we discuss that by such a solution, the reconstructed $\tilde{x}$ recovers the ground truth monosemantic feature x.

Specifically, when $W_m = (W_p, \mathbf{0})^\top$ and $S \to 1$, for a non-negative and non-zero x, we have with probability $nS^{n-1}(1-S)/(1-S^n) \to 1$ that there exists some $i \in [n]$ such that $x = x_i \boldsymbol{e}_i$, where $x_i > 0$. Correspondingly we have

$$\sigma(W_m W_p x_i \boldsymbol{e}_i) = \sigma\left(\begin{bmatrix} W_p^\top \\ \mathbf{0} \end{bmatrix} W_p x_i \boldsymbol{e}_i\right) = \begin{bmatrix} \sigma(x_i W_p^\top W_{p,[:,i]}) \\ \mathbf{0} \end{bmatrix}. \tag{10}$$

For ReLU activation, when the superposed dimensions have negative inferences as discussed in [6], i.e., $W_{p,[:,i]}^\top W_{p,[:,j]} \leq 0$ for $j \neq i$, we have $\text{ReLU}(x_i W_p^\top W_{p,[:,i]}) = x_i \boldsymbol{e}_i$, and correspondingly $\text{ReLU}(W_m W_p x_i \boldsymbol{e}_i) = (x_i \boldsymbol{e}_i^\top, \mathbf{0}^\top)^\top$. On the other hand, for a properly chosen k, we always have Top-$k(W_m W_p x_i \boldsymbol{e}_i) = (x_i \boldsymbol{e}_i^\top, \mathbf{0}^\top)^\top$ regardless of the inference values of the superposed dimensions. That is, sparse autoencoders with either ReLU or Top-k activation can recover the ground truth feature x with probability 1.

Moreover, in Theorem 2, we show that the solution to SAEs is unique.

Theorem 2 (Uniqueness). *Let $\mathcal{L}_{\text{SAE}}$ be defined in* (6). *If $n_m = n$ and $S \to 1$, then $W_m^* = W_p^\top$ is the unique solution to* $\arg\min_{W_m} \mathcal{L}_{\text{SAE}}(W_m; x_p)$.

Theorems 1 and 2 together show that SAE can be identifiable, i.e., the SAE encoder recovers the ground truth monosemantic feature x and the optimal solution to the loss function is unique. More precisely, the theorems show that extreme ground truth feature sparsity, sparse activation (ReLU or Top-k), and enough hidden dimensions n_m are sufficient conditions for identifiable SAE.

3.2.2 Necessary Conditions for SAE Full Recovery

In this part, we discuss that these sufficient conditions (extreme feature sparsity, sparse activation, and enough hidden dimension) are also necessary through several counterexamples.

Counterexample 1 shows that extreme sparsity of the ground truth feature x is necessary. If x is not entirely sparse, it can not be fully recovered by SAEs.

Counterexample 1 (SAEs without feature sparsity). *Suppose $n_m = n = 3$, $n_p = 2$, $x = (1, 1, 0)^\top$, and $W_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$. Then we have*

$$\sigma(W_m W_p x) = \sigma(W_p^\top W_p x) = \sigma\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \neq x. \tag{11}$$

Counterexample 2 shows that sparse activation (such that ReLU or Top-k) is necessary for identifiable SAEs. Without the sparse activation function, SAEs fail to recover the ground truth monosemantic features. Moreover, in Counterexample 3, we show that SAEs without sparse activation still fail even with sparse regularization such as the l_1 norm.

Counterexample 2 (SAEs without sparse activation). *Denote $\mathcal{L}'_{\text{SAE}} = \mathbb{E}_x \|x_p - W_m^\top W_m x_p\|^2$. Suppose $n_m = n = 3$, $n_p = 2$, $x = (0, 0, 1)^\top$, and $W_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$. Then $W_m = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \in \arg\min \mathcal{L}'_{\text{SAE}}$, but $W_m x_p = (0, -1, 0)^\top \neq x$.*

Counterexample 3 (SAEs without sparse activation with l_1 regularization). *Denote $\mathcal{L}''_{\text{SAE}} = \mathbb{E}_x \|x_p - W_m^\top W_m x_p\|_2 + \lambda \|W_m x_p\|_1$ for some $\lambda > 0$, where $\|\cdot\|_2$ and $\|\cdot\|_1$ denote the l_2 and l_1 norms, respectively. Suppose $n_m = n = 2$, $n_p = 1$, $x = (1, 0)^\top$, and $W_p = [1 \ -1]$. Then we have $W_m^* = \arg\min_{W_m} \mathcal{L}''_{\text{SAE}} = (\pm \min\{\lambda/2, \sqrt{2}/2\}, \pm \min\{\lambda/2, \sqrt{2}/2\})^\top$, and $W_m^* x_p = (\pm \min\{\lambda/2, \sqrt{2}/2\}, \pm \min\{\lambda/2, \sqrt{2}/2\})^\top \neq x$.*

Enough hidden dimensions in SAEs are also necessary for recovering the ground truth monosemantic features. Counterexample 4 shows that without enough hidden dimension, even recovering part of the ground truth monosemantic features is impossible.

Counterexample 4 (SAEs without enough hidden dimension). *Suppose $n = 3$, $n_m = n_p = 2$, $x = (0, 0, 1)^\top$, and $W_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$. Then if an SAE recovers part of the monosemantic dimensions, i.e., $\sigma(W_m^* x_p) = (x_1, x_2)^\top = (0, 0)^\top$, we should have $x_p = (W_m^*)^\top \sigma(W_m^* x_p)$. However, as $x_p = (0, -1)^\top \neq W_m (0, 0)^\top$ for all $W_m \in \mathbb{R}^{2 \times 2}$, we show that the SAE-recovered $\sigma(W_m x_p) \neq (x_1, x_2)^\top$ for all W_m.*

4 Improving Identifiability of SAEs with Adaptive Reweighting

In the previous section, we show that with SAEs, the ground truth monosemantic feature is fully recoverable under the three conditions. However, in practice, although the ground truth features are typically considered to be sparse [6], the sparsity level may not be to the extreme ($S \to 1$), making SAEs unidentifiable according to Section 3. Therefore, in this section, we aim to propose a possible improvement for the identifiability of SAEs regardless of the sparsity level S.

Specifically, in Section 4.1, we derive the loss gap between SAE reconstruction and ground truth reconstruction, where a non-zero gap leads to unidentifiability of an SAE. In Section 4.2, we discuss

that the gap can possibly be narrowed by leveraging an adaptive reweighting strategy, with the selection principle of weights discussed in Section 4.3. All proofs can be found in Appendix A.2.

4.1 The Gap between SAE Reconstruction and Ground Truth Reconstruction

To begin with, we observe that the SAE loss is not a direct reconstruction of the ground truth monosemantic feature x. Instead, it reconstructs the polysemantic x_p because the ground truth x is unknown. Therefore, we aim to compare the reconstruction loss of x and that of x_p through a loss decomposition.

Without loss of generality, we assume $n_m = n$, and define the reconstruction loss of the ground truth monosemantic feature x as

$$\mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p) = \mathbb{E}_x \|x - \mathrm{ReLU}(W_m x_p)\|^2 = \mathbb{E}_x \|x - \mathrm{ReLU}(W_m W_p x)\|^2. \tag{12}$$

Then we have the following loss decomposition.

Theorem 3 (Decomposition of $\mathcal{L}_{\mathrm{SAE}}$)**.** *Let $\mathcal{L}_{\mathrm{Rec}}$ be the reconstruction loss of the ground truth x defined in Equation* (12)*. Then the reweighted SAE loss $\mathcal{L}_{\mathrm{WSAE}}$ defined in Equation* (6) *has the following decomposition.*

$$\begin{aligned}\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) = \mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p) &+ \mathbb{E}_x \|(W_m - W_p^\top)^\top \mathrm{ReLU}(W_m W_p x)\|^2 \\ &+ 2\mathrm{ReLU}(W_m W_p x)^\top (W_m - W_p^\top) W_p [x - \mathrm{ReLU}(W_m W_p x)] \\ &+ [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{13}$$

Theorem 4 shows the gap between the SAE loss $\mathcal{L}_{\mathrm{SAE}}(W_m; x_p)$ and the reconstruction loss of the ground truth features $\mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p)$. By Theorem 1, under the extreme sparsity condition, if $W_m = W_p^\top$, the gap turns 0, and then the reconstruction of the polysemantic x_p is equivalent to the reconstruction of the ground truth x, i.e., the SAE is identifiable.

Otherwise, the gap between $\mathcal{L}_{\mathrm{SAE}}(W_m; x_p)$ and $\mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p)$ is composed of three terms. The first two terms measure the difference between W_m and $W_p^\top$. When $W_m = W_p^\top$, the first two terms equal 0. The last term measures the difference between $W_p^\top W_p$ and the identity matrix $I_{n\times n}$, which cannot be reduced because in SAEs, W_p is given as an input rather than learned.

4.2 Adaptively Reweighted Sparse Autoencoders

To narrow the gap, we propose the reweighted SAEs with adaptive weights according to the polysemantic level of each dimension, where the adaptive weights do not essentially affect the first two gap terms but can help reduce the third one. Specifically, we introduce a weight matrix $\Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_{n_p})$, and propose the re-Weighted SAE (WSAE) loss as

$$\mathcal{L}_{\mathrm{WSAE}}(W_m; x_p) = \mathbb{E}_{x_p} \|\Gamma[x_p - W_m^\top \mathrm{ReLU}(W_m x_p)]\|_2^2. \tag{14}$$

Theorem 4 (Decomposition of $\mathcal{L}_{\mathrm{WSAE}}$)**.** *Let $\mathcal{L}_{\mathrm{Rec}}$ be the reconstruction loss of the ground truth x defined in Equation* (12)*. Then the reweighted SAE loss $\mathcal{L}_{\mathrm{WSAE}}$ defined in Equation* (14) *has the following decomposition.*

$$\begin{aligned}&\mathcal{L}_{\mathrm{WSAE}}(W_m; x_p) \\ &= \mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p) + \mathbb{E}_x \|\Gamma(W_m - W_p^\top)^\top \mathrm{ReLU}(W_m W_p x)\|^2 \\ &\quad + 2\mathrm{ReLU}(W_m W_p x)^\top (W_m - W_p^\top) \Gamma^\top \Gamma W_p [x - \mathrm{ReLU}(W_m W_p x)] \\ &\quad + [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top \Gamma^\top \Gamma W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{15}$$

Theorem 4 shows the gap between a reweighted SAE loss $\mathcal{L}_{\mathrm{SAE}}(W_m; x_p)$ and the reconstruction loss of the ground truth features $\mathcal{L}_{\mathrm{Rec}}(W_m; x, x_p)$. As a special case, when the weight matrix Γ is an identity matrix (uniform weights), the decomposition of WSAEs in Theorem 4 degenerates to that of SAEs shown in Theorem 3.

By Theorem 4, we note that the diagonal weight matrix Γ does not affect the optimum of the first two gap terms, i.e., the first two gap terms still equal 0 when $W_m = W_p^\top$. Nonetheless, it changes the

optimum of the last gap term, i.e., the last gap terms equal 0 when $W_p^\top \Gamma^\top \Gamma W_p = I_{n\times n}$. Compared with the last term in Equation (13), the last gap term in Equation (15) can be effectively narrowed by selecting a proper weight matrix Γ. That is, compared with SAEs with uniform weights, WSAEs with adaptive weights can lead to better reconstruction of the ground truth monosemantic feature x, and correspondingly have better identifiability.

4.3 Principle of Weight Selection

In this part, we discuss how to select a proper Γ to narrow the last gap term in Equation (15). Specifically, because Γ is a diagonal matrix, we have

$$W_p^\top \Gamma^\top \Gamma W_p = W_p^\top \Gamma^2 W_p = \begin{bmatrix} \gamma_1^2 \|W_{p,[:,1]}\|^2 & \cdots & \gamma_1^2 W_{p,[:,1]}^\top W_{p,[:,n]} \\ \cdots & & \\ \gamma_n^2 W_{p,[:,n]}^\top W_{p,[:,1]} & \cdots & \gamma_n^2 \|W_{p,[:,n]}\|^2. \end{bmatrix} \tag{16}$$

By taking a closer look at the polysemantic feature x_p, we focus on the polysemantic level of each dimension. Without loss of generality, for $0 < i < j < n_p$, we assume x_j has an equal or higher polysemantic level than x_i. From the perspective of the weight matrix W_p, this means there are more negative values in $\{W_{p,[:,j]}^\top W_{p,[:,k]}\}_{k=1}^{n_p}$ than in $\{W_{p,[:,i]}^\top W_{p,[:,k]}\}_{k=1}^{n_p}$. Therefore, in Equation (16), we assign smaller weights to the inference terms related to $W_{p,[:,j]}$, reducing the effect of the off-diagonal negative inferences in Equation (16), i.e., we let $\gamma_j < \gamma_i$. In other words, for the reconstruction of x_p, we assign larger weights to the monosemantic dimensions and assign smaller weights to the polysemantic dimensions.

For better understanding, we illustrate this selection principle in Example 1.

Example 1. *Suppose $n = 3$, $n_p = 2$, and $W_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$. Then $x_p = W_p x = (x_1, x_2 - x_3)^\top$, where the first dimension is monosemantic, and the second dimension is polysemantic. Moreover, we have $W_{p,[:,1]}^\top W_{p,[:,2]} = W_{p,[:,1]}^\top W_{p,[:,3]} = 0$, whereas $W_{p,[:,2]}^\top W_{p,[:,3]} = -1$. By Equation (16), to reduce the negative inference between $W_{p,[:,2]}$ and $W_{p,[:,3]}$, we need to reduce the weight on the second dimension γ_2.*

5 Validation Experiments

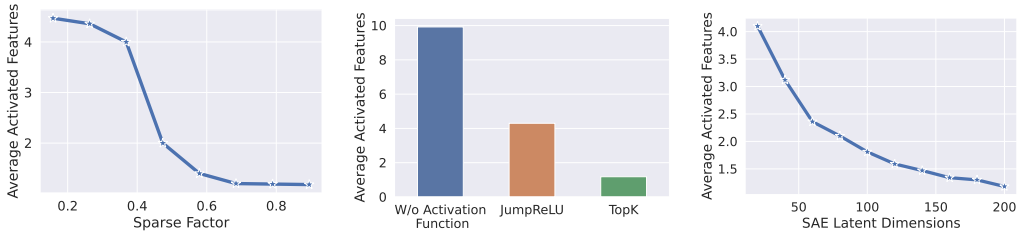
In this section, we conduct numerical experiments to validate our theoretical findings. Specifically, in Section 5.1, we verify the necessary and sufficient conditions for identifiable SAEs discussed in Section 3.2 through experiments on synthetic data. In Section 5.2, we validate the effectiveness of our proposed reweighting strategy by evaluating the reweighted SAEs on real-world datasets and pretrained models in both language and vision domains.

5.1 Validation of the Necessary and Sufficient Conditions

In Section 3.2.1 and 3.2.2, we provide a theoretical analysis of the sufficient and necessary conditions for SAEs to recover ground truth monosemantic features. In this section, we will empirically validate these conditions with the toy models introduced in Section 3.1.

Setups. We follow the settings in Section 3.1 and examine three conditions: (1) input sparsity, (2) the role of activation functions in SAE latents, and (3) the importance of having a sufficiently large hidden dimension. Specifically, we set $n = 200$ and $n_p = 20$ to generate polysemantic embeddings and then attempt to extract monosemantic features with SAEs trained on frozen polysemantic embeddings. When analyzing the influence of input sparsity, we set $x_i = 0$ with a probability $S \in [0, 1]$ for $i \in [n]$. To assess the necessity of activation functions, we compare three variants: SAEs without activation functions, SAEs using JumpReLU (with a threshold of 0.1), and SAEs using Top-k selection (with k=2). For the necessity of enough hidden dimensions, we vary the hidden dimension of SAEs from 20 to 200. While varying one of the conditions, we keep the others as their optimal choice.

During evaluation, we measure the monosemanticity of SAE latents as the metric. For a sample $x \in \mathbb{R}^n$, we perform n encodings, each time isolating a single input feature. Specifically, for the i-th encoding, we construct $x^i = (0, \cdots, x_i, \cdots, 0)$, where only the i-th feature is retained. This yields



(a) Input Sparsity (b) Activation Functions (c) SAE Latent Dimensions

Figure 2: The empirical verifications of the necessary conditions for SAE recovery of monosemantic features on the toy model ($n = 200$, $n_p = 20$). We use the average number of ground-truth features that are activated in the same SAE dimension to evaluate the monosemanticity of SAE latents, where fewer average activated features indicate stronger monosemanticity.

n SAE latent representations $\{h(x^i)\}_{i=1}^n$, each corresponding to the activation induced by a single input feature. We then calculate the total activation values across different samples in each SAE latent dimension. For example, for j-th dimension in SAE latents, we define $M_j^i = \sum_x h_j(x^i)$ representing the cumulative activation from the i-th input feature. If the value M_j^i exceeds a threshold, we consider the j-th SAE latent to be activated by the i-th input. Finally, we compute the average number of input features that activate each SAE latent dimension. We note that if the SAE successfully recovers monosemantic features, each latent should be activated by only a small number of input features.

Results. As shown in Figure 2, we observe a clear decrease in the average number of activated features (indicating a significant improvement in the monosemanticity) under three conditions: increased input sparsity, the use of activation functions, and higher latent dimensionality in SAE latents. These empirical findings align well with our theoretical analysis, confirming that the necessary and sufficient conditions for identifiable SAEs are: (1) sparsity in the ground truth features, (2) sparse activation in the SAE latent space, and (3) a sufficiently large number of hidden dimensions in the SAE.

5.2 Empirical Evaluation of Reweighted SAEs

5.2.1 Experiments on Language Models

In this section, we evaluate the effectiveness of our proposed strategies on real-world datasets and pretrained language models.

Setups. We use Pythia-160M [1] as the frozen language model backbone and train sparse autoencoders (SAEs) on its internal activations. The latent dimension of the SAE is set to 32 times the input dimension, and we employ Top-k activation ($k = 32$) in the latent layer of SAEs. Guided by the theoretical analysis, we compare two training objectives: a standard reconstruction loss with uniform weights, and a weighted reconstruction loss that emphasizes monosemantic features.

Uunlike in toy models, we cannot directly access the ground-truth monosemanticity of features in real-world settings. However, several surrogate metrics have been proposed to evaluate monosemanticity. For instance, [23] use variance within each feature dimension as a surrogate while [17] introduce the auto-interpretability score, which leverages large language models to summarize whether the features activated in the same dimension are semantically similar. In our experiments, we use the variance as the metric when assigning weights, due to its computational simplicity. Specifically, let s_i denote the variance of activations in SAE latent dimension i. We set the weight matrix in Equation (14) as

$$\Gamma = \text{diag}(s_1^\alpha, \cdots, s_{n_p}^\alpha). \tag{17}$$

As [23] suggest that higher variance indicates stronger monosemanticity, when α increases, the reconstruction loss assigns more weights to the reconstruction of monosemantic features.

While for evaluation of SAE latents, we use the auto-interpretability score [17] as it is more accurate to assess monosemanticity. To compute the score for a given SAE latent dimension, we first identify the top-activated samples for that dimension. A large language model (e.g., Llama3.1-8B [22]) is then used to generate a natural language summary describing the shared characteristics of these samples. In the next step, another language model is prompted with the generated summary to predict whether additional samples in the dataset would activate the given SAE dimension. The prediction accuracy serves as the auto-interpretability score: higher accuracy indicates that the activated samples in the

Table 1: Auto-interpretability scores (%) of SAEs trained following different layers (0-11) of Pythia-160M with original SAE and weighted SAE loss. SAEs trained with weighted SAE loss obtain higher auto-interpretability scores (i.e., stronger monosemanticity) across different situations.

	0	1	2	3	4	5	6	7	8	9	10	11
Original SAE	74.7	74.1	76.7	77.8	78.5	79.5	79.3	77.8	74.6	75.6	71.6	72.5
Weighted SAE (α=0.5)	75.4	77.6	76.4	77.9	79.3	79.6	79.8	77.4	78.6	**79.3**	**76.1**	72.6
Gains	+0.7	+3.5	-0.3	+0.1	+0.8	+0.1	+0.5	-0.4	+4.0	**+3.7**	**+4.5**	+0.1
Weighted SAE (α=1)	**77.2**	**78.9**	**81.3**	**84.6**	**83.9**	**83.3**	**83.9**	**79.6**	**81.5**	77.6	72.4	**73.5**
Gains	**+2.5**	**+4.8**	**+4.6**	**+6.8**	**+5.4**	**+3.8**	**+4.6**	**+1.8**	**+6.9**	+2.0	+0.8	+1.0

dimension are semantically similar, reflecting stronger monosemanticity. More details can be found in Appendix B.

Results. As shown in Table 1, assigning greater weight to the reconstruction of monosemantic features leads to a significant improvement in the monosemanticity of SAE latents, yielding an average 3.8% gain in auto-interpretability score when $\alpha = 1$. The improvements is consistent across SAEs trained on different layers of backbone models. We also note that the improvement is more significant when the baseline SAE already exhibits relatively strong monosemanticity. These empirical findings further support our theoretical insight: prioritizing monosemantic features during training SAEs enhances the monosemanticity of the learned SAE latents

5.2.2 Experiments on Vision Models

In addition to language models, we also evaluate our strategy on SAEs trained following pretrained vision backbones. As noted by [24], features learned by current vision pertaining paradigms, such as contrastive learning, are often highly polysemantic, making it difficult to select monosemantic dimensions. Consequently, we adopt Non-negative Contrastive Learning (NCL) [24] to pretrain the bakcbone, as it can simultaneously obtain monosemantic and polysemantic features.

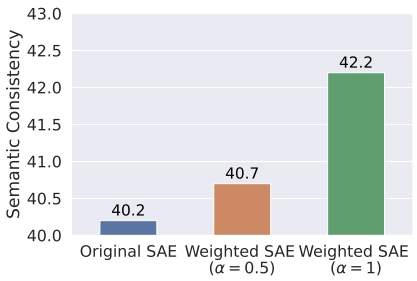


Figure 3: Semantic consistency (%) of SAEs trained on the embeddings of ResNet-18 with original SAE and weighted SAE loss.

Specifically, we pretrain a ResNet-18 with NCL on ImageNet-100, and subsequently train SAEs on the learned representations. During the training process of SAEs, we set the latent dimension to 16384 and use the topK sparse autoencoder with $k = 16$. When evaluating monosemanticity, we follow [24] and use semantic consistency as the metric, which calculates the proportion of top-activated samples that belong to their most frequent class along each dimension. As the higher semantic consistency indicates stronger monosemanticity, we respectively train SAEs with uniform weights and more weights on the dimension with higher semantic consistency. Specifically, we set the weight matrix in Equation (14) as $\Gamma = \text{diag}(\beta_1^\alpha, \cdots, \beta_{n_p}^\alpha)$, where β_i is the semantic consistency of the i-th dimension. In Figure 3, we observe that assigning greater weight to the reconstruction of monosemantic features leads to a notable improvement in semantic consistency (monosemanticity) of SAE latents. These results further validate both our theoretical analysis and the effectiveness of the proposed weighting strategy.

6 Conclusion & Limitations

In this paper, we proposed a theoretical framework for identifying the conditions under which sparse autoencoders (SAEs) can accurately recover ground truth monosemantic features from superposed polysemantic representations. We established that extreme sparsity of the ground truth features, sparse activation of the SAE, and a sufficient hidden dimension are necessary and sufficient for identifiability. To address cases where these conditions are not fully met, we introduced a reweighting strategy that significantly improves feature monosemanticity and interpretability. Experimental results validated our theoretical insights, demonstrating that our weighted SAE outperforms uniformly weighted models in recovering interpretable features.

Although this paper has the limitation that we only test the monosemanticity of ReLU and Top-k activated SAEs, without verifying their effectiveness in further applications such as detecting harmful neurons in large language models, our work still offers practical insights for enhancing feature interpretability. We leave the applications of our proposed reweighted SAEs with better identifiability to further NLP, vision, and multi-modal tasks for future works.

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A Proofs

A.1 Proofs Related to Section 3

Proof of Theorem 1. When $W_m = (W_p, \mathbf{0})^\top$, we have

$$\begin{aligned}\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) &= \mathbb{E}_x \|W_p x - (W_p, \mathbf{0})\sigma(\begin{bmatrix} W_p^\top \\ \mathbf{0} \end{bmatrix} W_p x)\|^2 \\ &= \mathbb{E}_x \|W_p x - W_p^\top \sigma(W_p^\top W_p x)\|^2. \end{aligned} \tag{18}$$

When $S \to 1$, we have

$$\begin{aligned}&\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) \\ &= \sum_{u=1}^n S^n \mathcal{L}(W_m; W_p, x = \mathbf{0}) + \sum_{u=1}^n \mathbb{E}_{x_u} C_n^1 S^{n-1}(1-S)\mathcal{L}(W_m; W_p, x = x_u \boldsymbol{e}_u) + o(1-S) \\ &= 0 + nS^{n-1}(1-S)\sum_{u=1}^n \mathbb{E}_{x_u}\mathcal{L}(W_m; W_p, x = x_u \boldsymbol{e}_u) + o(1-S) \\ &= nS^{n-1}(1-S)\sum_{u=1}^n \mathbb{E}_{x_u}\mathcal{L}(W_m; W_p, x = x_u \boldsymbol{e}_u) \\ &= nS^{n-1}(1-S)\sum_{u=1}^n \mathbb{E}_{x_u}\|W_{p,[:,u]} - W_p^\top \sigma(W_p^\top W_{p,[:,u]})\|^2. \end{aligned} \tag{19}$$

For ReLU activation, when the superposed dimensions have negative inferences as discussed in [6], i.e., $W_{p,[:,i]}^\top W_{p,[:,j]} \le 0$ for $i \ne j$, we have $\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) = 0 = \min_{W_m} \mathcal{L}_{\mathrm{SAE}}(W_m; x_p)$.

On the other hand, for top-k activation, when $1 < k \le n_m - p$, where $p := \max_{i\in[n]} |\{j \in [n] : W_{p,[:,i]}^\top W_{p,[:,j]} < 0\}|$, we have Top-$k(W_p^\top W_{p,[:,u]}) = \boldsymbol{e}_u$, and correspondingly $\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) = 0 = \min_{W_m} \mathcal{L}_{\mathrm{SAE}}(W_m; x_p)$. □

Proof of Theorem 2. Note that Top-$k(x) = \mathrm{ReLU}(x - x_{(k+1)}\mathbf{1}) + x_{(k+1)}\mathrm{H}(x - x_{(k+1)}\mathbf{1})$, where $H(x) := \mathbf{1}[x \ge 0]$ denotes the Heaviside function, $x_{(k+1)}$ denotes the $(k+1)$-th largest element of x, and $\mathbf{1}$ denotes the all-one vector. We consider the sparse activation function of the form $\sigma(x) = \mathrm{ReLU}(x - c) + b$ to represent both ReLU and Top-k, having the gradient $\frac{\partial \sigma(x)}{\partial x} = \mathrm{H}(x - c)$. Then by definition, we have

$$\begin{aligned}&\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) \\ &= \mathbb{E}_x \|W_p x - W_m^\top \sigma(W_m W_p x)\|_2^2 \\ &= \mathbb{E}_x [W_p x - W_m^\top \sigma(W_m W_p x)]^\top [W_p x - W_m^\top \sigma(W_m W_p x)] \\ &= \mathbb{E}_x x^\top W_p^\top W_p x - 2x^\top W_p^\top W_m^\top \sigma(W_m W_p x) \\ &\quad + \sigma(W_m W_p x)^\top W_m W_m^\top \sigma(W_m W_p x) \\ &= \mathbb{E}_x x^\top W_p^\top W_p x - 2\sum_{q=1}^{n_p}\sum_{j=1}^{n_m} x^\top W_{p,[q,:]}^\top W_{m,[j,q]} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,:]} x) \\ &\quad + \sum_{j=1}^{n_m}\sum_{q=1}^{n_p} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,:]} x) W_{m,[j,q]} \sum_{s=1}^{n_m} W_{m,[s,q]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,:]} x). \end{aligned} \tag{20}$$

If $S \to 1$, by (19), we have

$$\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) = nS^{n-1}(1-S)\sum_{u=1}^n \mathbb{E}_{x_u}\mathcal{L}_u(W_m; W_p, x_u), \tag{21}$$

where for $u \in [n]$ and $x_u > 0$, by (20), we denote

$$\mathcal{L}_u(W_m; W_p, x_u)$$

$$:= x_u^2 W_{p,[:,u]}^\top W_{p,[:,u]} - 2x_u^2 \sum_{q=1}^{n_p}\sum_{j=1}^{n_m} W_{p,[q,u]} W_{m,[j,q]} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,u]})$$
$$+ x_u^2 \sum_{j=1}^{n_m}\sum_{q=1}^{n_p} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,u]}) W_{m,[j,q]} \sum_{s=1}^{n_m} W_{m,[s,q]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]}). \tag{22}$$

Then for $k \in [n_m]$ and $l \in [n_p]$, by (19), we have

$$\frac{\partial \mathcal{L}_u(W_m; W_p, x_u)}{\partial W_{m,[k,l]}}$$
$$= -2x_u^2 W_{p,[l,u]} \sigma(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]})$$
$$- 2x_u^2 \sum_{q=1}^{n_p} W_{p,[q,u]} W_{m,[k,q]} \mathrm{H}(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]} - c) W_{p,[l,u]}$$
$$+ x_u^2 \sum_{q=1}^{n_p} \mathrm{H}(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]} - c) W_{p,[l,u]} W_{m,[k,q]} \sum_{s=1}^{n_m} W_{m,[s,q]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]})$$
$$+ x_u^2 \sigma(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]}) \sum_{s=1}^{n_m} W_{m,[s,l]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]})$$
$$+ x_u^2 \sum_{j=1}^{n_m} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,u]}) W_{m,[j,l]} \sigma(\sum_{t=1}^{n_p} W_{m,[k,t]} W_{p,[t,u]})$$
$$+ x_u^2 \sum_{j=1}^{n_m}\sum_{q=1}^{n_p} \sigma(\sum_{i=1}^{n_p} W_{m,[j,i]} W_{p,[i,u]}) W_{m,[j,q]} W_{m,[k,q]} \mathrm{H}(\sum_{t=1}^{n_p} W_{m,[k,t]} W_{p,[t,u]} - c) W_{p,[l,u]}$$
$$= 2x_u^2 \sum_{q=1}^{n_p} \mathrm{H}(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]} - c) W_{p,[l,u]} W_{m,[k,q]}$$
$$\cdot \Big[\sum_{s=1}^{n_m} W_{m,[s,q]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]}) - W_{p,[q,u]} \Big]$$
$$+ 2x_u^2 \sigma(\sum_{i=1}^{n_p} W_{m,[k,i]} W_{p,[i,u]}) \Big[\sum_{s=1}^{n_m} W_{m,[s,l]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]}) - W_{p,[l,u]} \Big]. \tag{23}$$

Observe that $\frac{\partial \mathcal{L}_u(W_m; W_p, x_u)}{\partial W_{m,[k,l]}} = 0$ for $k \in [n_m]$ and $l \in [n_p]$ holds if and only if

$$\sum_{s=1}^{n_m} W_{m,[s,l]} \sigma(\sum_{t=1}^{n_p} W_{m,[s,t]} W_{p,[t,u]}) = W_{p,[l,u]} \tag{24}$$

for $l \in [n_p]$, or equivalently

$$W_{p,[:,u]} = W_m^\top \sigma(W_m W_{p,[:,u]}). \tag{25}$$

For ReLU activation, the optimal condition becomes

$$W_p = W_m^\top \mathrm{ReLU}(W_m W_p), \tag{26}$$

When the superposed dimensions have negative inferences as discussed in [6], i.e., $W_{p,[:,i]}^\top W_{p,[:,j]} \leq 0$ for $i \neq j$, if $n_m = n$, the unique solution is $W_m = W_p^\top$.

On the other hand, for top-k activation, the optimal condition becomes

$$W_{p,[:,u]} = W_m^\top [\mathrm{ReLU}(W_m W_{p,[:,u]} - c_u \mathbf{1}) + c_u \mathrm{H}(W_m W_{p,[:,u]} - c_u \mathbf{1})]. \tag{27}$$

If $0 < c_u < \|W_{p,[:,u]}\|^2$ and $n_m = n$, the unique solution is $W_m = W_p^\top$. When $x \in [0,1]$, as $c_u = (W_p(x_u \boldsymbol{e}_u))_{(k)} = (x_u W_{p,[:,u]})_{(k)}$, this is equivalent to having $0 < (W_{p,[:,u]})_{(k)} < \|W_{p,[:,u]}\|^2$, which can be guaranteed by having $1 < k \leq n_m - p$, where $p := \max_{i \in [n]} |\{j \in [n] : W_{p,[:,i]}^\top W_{p,[:,j]} < 0\}|$. □

A.2 Proofs Related to Section 4

Proof of Theorem 3. By definition, we have

$$\begin{aligned}\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) &= \mathbb{E}_x \|W_p x - W_m^\top \mathrm{ReLU}(W_m W_p x)\|^2 \\ &= \mathbb{E}_x \|W_p[x - \mathrm{ReLU}(W_m W_p x)] + (W_p - W_m^\top)\mathrm{ReLU}(W_m W_p x)\|^2 \\ &= \mathbb{E}_x \|W_p[x - \mathrm{ReLU}(W_m W_p x)]\|^2 + \|(W_m - W_p^\top)^\top \mathrm{ReLU}(W_m W_p x)\|^2 \\ &\quad + 2\mathrm{ReLU}(W_m W_p x)^\top (W_m - W_p^\top) W_p[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{28}$$

The first term of (30) can be further decomposed into

$$\begin{aligned}&\|W_p[x - \mathrm{ReLU}(W_m W_p x)]\|^2 \\ &= [x - \mathrm{ReLU}(W_m W_p x)]^\top W_p^\top W_p [x - \mathrm{ReLU}(W_m W_p x)] \\ &= [x - \mathrm{ReLU}(W_m W_p x)]^\top [x - \mathrm{ReLU}(W_m W_p x)] \\ &\quad + [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)] \\ &= \|x - \mathrm{ReLU}(W_m W_p x)\|^2 \\ &\quad + [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{29}$$

Combining (30) and (31) finishes the proof. □

Proof of Theorem 4. By definition, we have

$$\begin{aligned}\mathcal{L}_{\mathrm{SAE}}(W_m; x_p) &= \mathbb{E}_x \|\Gamma[W_p x - W_m^\top \mathrm{ReLU}(W_m W_p x)]\|^2 \\ &= \mathbb{E}_x \|\Gamma W_p[x - \mathrm{ReLU}(W_m W_p x)] + \Gamma(W_p - W_m^\top)\mathrm{ReLU}(W_m W_p x)\|^2 \\ &= \mathbb{E}_x \|\Gamma W_p[x - \mathrm{ReLU}(W_m W_p x)]\|^2 + \|\Gamma(W_m - W_p^\top)^\top \mathrm{ReLU}(W_m W_p x)\|^2 \\ &\quad + 2\mathrm{ReLU}(W_m W_p x)^\top (W_m - W_p^\top)\Gamma^\top \Gamma W_p[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{30}$$

The first term of (30) can be further decomposed into

$$\begin{aligned}&\|\Gamma W_p[x - \mathrm{ReLU}(W_m W_p x)]\|^2 \\ &= [x - \mathrm{ReLU}(W_m W_p x)]^\top W_p^\top \Gamma^\top \Gamma W_p [x - \mathrm{ReLU}(W_m W_p x)] \\ &= [x - \mathrm{ReLU}(W_m W_p x)]^\top [x - \mathrm{ReLU}(W_m W_p x)] \\ &\quad + [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top \Gamma^\top \Gamma W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)] \\ &= \|x - \mathrm{ReLU}(W_m W_p x)\|^2 \\ &\quad + [x - \mathrm{ReLU}(W_m W_p x)]^\top (W_p^\top \Gamma^\top \Gamma W_p - I_{n\times n})[x - \mathrm{ReLU}(W_m W_p x)]. \end{aligned} \tag{31}$$

Combining (30) and (31) finishes the proof. □

B Experiments Details

B.1 Compute resources

For the experiments on toy models in Section 3, we conduct experiments on an NVIDIA 3090 GPU with 24GB of memory. The training and evaluation of a toy model takes around 30 minutes. For the experiments on language models in Section 5.1, we conduct experiments on an NVIDIA A100 GPU with 40GB of memory. The training of a sparse autoencoder takes around 24 hours, and the evaluation needs 10 minutes. While for experiments on vision models in Section 5.2, we conduct experiments on two NVIDIA 3090 GPUs with 24GB of memory. The training of a sparse autoencoder takes around 12 hours, and the evaluation needs 5 minutes.